CURRICULUM VITAE

Qian Ouyang

Machine learning for materials and molecular simulation.

qouyang@nd.edu | cv.qianoy.uk

github.com/Owen-Ou-Yang

Education

Xi'an Jiaotong University 2023 - 2027 (expected)

Undergraduate studies | Artificial Intelligence

Artificial Intelligence Experimental Class

University of Notre Dame August - December 2025

Non-degree Exchange Student (undergraduate) | Fall 2025

Semester GPA: **3.917/4.0**, 12 credits

Selected coursework: Computer Vision - graduate level (A); Natural Language Processing (A); Computer Graphics (A).

Academic experience

Research Assistant October 2025 - Present

University of Notre Dame

Prof. Tengfei Luo's group | Department of Aerospace and Mechanical Engineering

Summer Research Participant (iSURE) July - September 2026

University of Notre Dame

International Summer Undergraduate Research Experience (iSURE).

Selected research

Molecular representations and physical prediction

Encoder comparisons, representation ablations, and physical-readout evaluation, from earlier CG polymer and MACE water studies to current pretrained-feature transfer research.

Pretrained atomistic representations: transfer and label efficiency

Ongoing research · crystal study in preparation

Compared pretrained and randomly initialized Atom-JEPA encoders on QM9 molecular polarizability prediction under matched data and downstream update budgets. Examined how pretraining effects varied with optimization seeds and checkpoint selection. A crystal-property extension is in preparation.

State-space models for controlled physical response

Exploratory study completed | Closed 24 October 2026

Compared state-space and recurrent models on a synthetic Bouc-Wen system. Evaluated state/control adapters, functional readouts, and symmetry constraints against full-curve and paired control-response targets.

Selected research (continued)

MLIP material properties and efficiency

Polymer-density workflow development and HPC resource evaluation, building on earlier UMA / FAIR-Chem workflow exploration.

Polymer density workflows with MACE

Ongoing workflow development | Displayed pilot: 17 September 2026

Developed polymer-density workflows connecting RadonPy/GAFF2 preparation with MACE molecular dynamics and reference comparisons. Ran preliminary simulations, checked chemical identity and sampling quality, and analyzed reference matching and HPC resource requirements; workflow development continues.

GPU resource trade-offs in MACE molecular dynamics

Engineering pilot | September 2026

Ran a fixed MACE polymer workload with two- and four-GPU configurations, measuring wall time, sampled device memory, and allocated GPU-hours. Analyzed the runtime-memory-cost trade-off in a single-run engineering pilot on different hosts.

Polymer elastic-response workflows with UMA / FAIR-Chem

Resource-limited exploration

Prepared UMA / FAIR-Chem-LAMMPS workflow components for finite-strain stress sampling, batch execution, and graph inspection. GPU-memory constraints limited the work before validated property results and motivated subsequent MLIP resource studies.

Selected course project

Driving Coach: Lane and Traffic-Object Detection

Computer Vision, University of Notre Dame | Fall 2025

Instructor: Prof. Adam Czajka

- Adapted a YOLOv1-style detector with a pretrained ResNet-18 backbone for BDD100K driving scenes, including lane and traffic-object detection.
- Investigated lane-representation and training issues, and qualitatively evaluated the models on an unseen Shenzhen highway video, analyzing failures involving reflections, lane geometry, and traffic signs.

Technical skills

Programming: Python, C++

Machine Learning: PyTorch, torchvision, TensorBoard.

Scientific Computing / Research Workflows: NumPy, SciPy, Matplotlib; LAMMPS, ASE, MACE, UMA / FAIR-Chem; CRC/SGE batch workflows and GPU resource measurement.